

Technology Stack

Date	30 June 2026
Team ID	*****
Project Name	Smart Lender – AI-Powered Loan Eligibility Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

The AI-Powered Loan Eligibility Prediction System uses modern technologies to develop an intelligent and user-friendly web application. Python is used as the core programming language, while Flask handles the backend functionality. HTML, CSS, and JavaScript are used to create a responsive user interface. The machine learning model is developed using the Random Forest algorithm with the help of Scikit-learn, Pandas, and NumPy libraries. The trained model is stored as a Pickle (.pkl) file for efficient prediction. GitHub is used for version control and project management, ensuring easy maintenance and collaboration.

Technology Stack Details

S. No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., HTML, CSS & JavaScript)	HTML5, CSS3, JavaScript	Used to create a responsive and user-friendly web interface where applicants enter loan details and view prediction results.
2	Backend / Server-Side (e.g., Python, Flask)	Python and Flask	Flask integrates seamlessly with the Python-based ML stack (scikit-learn, XGBoost), allowing the trained model to be loaded and served for real-time predictions through simple REST routes.
3	Database / Data Storage (e.g., CSV Dataset, Pickle Model File)	CSV Dataset, Pickle (.pkl) Model File	The CSV dataset is used for training the model, while the trained Random Forest model is stored as a .pkl file for fast prediction..
4	Version Control & CI/CD (e.g., GitHub)	GitHub	GitHub enables seamless collaboration between the team (Yaswanth Ganesh R, Sandhya Mondam, Salma Shaik) for tracking code changes, managing branches for model experiments, and coordinating contributions.

S. No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
5	ML Framework & Libraries <i>(e.g., Scikit-learn, pandas, NumPy, Matplotlib, Joblib)</i>	Scikit-learn, Pandas, NumPy, Matplotlib, Joblib	Pandas and NumPy are used for data preprocessing, Scikit-learn for model training, Joblib for saving/loading the trained model, and Matplotlib for data visualization and analysis.